

Chapter 6

Object Reconstruction and Identification

Hadronically decaying τ -leptons, electrons, muons and jets are used in this analysis. This chapter describes their reconstruction, identification and calibration, followed by their definitions in the analysis, overlap removal and the reconstruction of E_T^{miss} .

These objects are reconstructed and identified using detector measurements. Tracks are used to determine charged-particle momenta and production vertices, while energy deposits are measured in the calorimeters. The reconstructed energies and momenta are calibrated to correct biases in the detector response. Simulated event weights are further corrected using efficiency scale factors.

6.1 Tracks

6.1.1 Reconstruction

A track is reconstructed by fitting the trajectory of a charged particle to measurements in the ID. Its momentum and charge are determined from the curvature in the magnetic field, while its position near the beam line is used to associate it with a production vertex. Track reconstruction is performed in the following stages [1]:

1. **Space-point formation.** Signals in neighbouring channels of a silicon sensor are grouped into clusters. The coordinates provided by Pixel clusters and SCT clusters are converted into three-dimensional space points.
2. **Seeding and track finding.** Combinations of space points are used to form track seeds. These seeds are extended through the silicon layers with a combinatorial Kalman filter, which is used to refine the track parameters.

3. **Ambiguity resolution.** Track candidates are ranked based on their fit quality, assigned clusters and other criteria. And candidates that share the same clusters are marked as "shared hits". At last the low-quality candidates that fail the track-quality or hit-sharing requirements are rejected to suppress duplicate and fake tracks.
4. **TRT extension.** Tracks are extended with additional TRT hits. The extended track is retained if its quality score is higher than that of the original track reconstructed using only pixel and SCT hits.

6.1.2 Track parameters and quality

The fitted trajectory is described by five parameters, as illustrated in Figure 6.1:

- d_0 : distance in the $x - y$ plane between the beam line and the perigee point, defined as the point on the track closest to the beam line.
- z_0 : the z coordinate of the perigee point relative to the reference plane at $z = 0$.
- ϕ and θ : the azimuthal and polar angles of the momentum, respectively.
- q/p : the electric charge divided by the momentum magnitude.

Two quantities are used to test whether a track is consistent with production at the primary vertex [2]:

- **Transverse impact-parameter significance**, d_0/σ_{d_0} : the transverse distance from the beam line in units of its uncertainty σ_{d_0} .
- **Projected longitudinal distance**, $\Delta z_0 \sin \theta = (z_0 - z_{PV}) \sin \theta$: the longitudinal distance between the track perigee and the primary vertex. Here, z_{PV} is the z coordinate of the primary vertex.

Small absolute values of both quantities suggest that the track is from the primary vertex. A large longitudinal distance can also indicate a track from another pp collision. Prompt electron and muon candidates are therefore selected by requiring small values of both quantities, and displaced candidates with large values can originate from a separate decay vertex, for example from a b - or c -hadron decay.

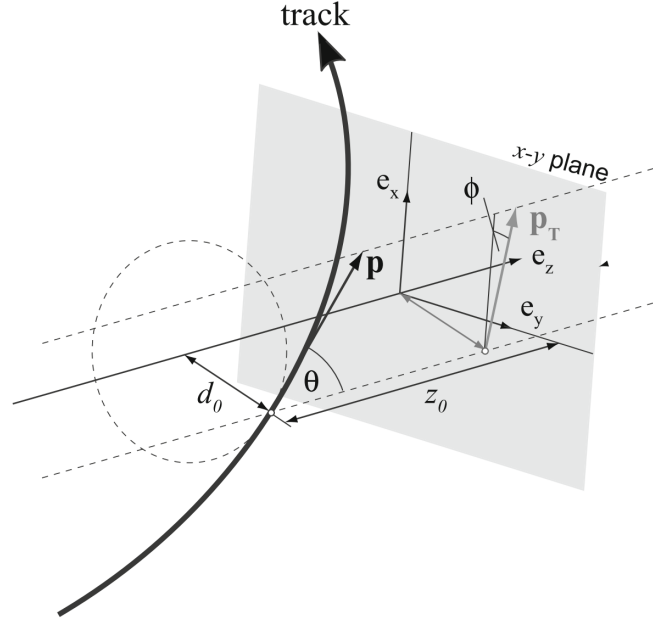


Fig. 6.1 Geometry of the Track parameters. The grey plane represents the reference $z = 0$ plane [3].

6.2 Primary vertex

A primary vertex (PV) is the reconstructed position of a pp collision. It is used as the reference for prompt-lepton selection, isolation, pile-up jet rejection and the track contribution to E_T^{miss} . Its position is determined from tracks originating from a common point. Several PVs can be reconstructed in an event because of pile-up. The vertex with the largest sum of squared transverse momenta of its associated tracks is selected as the PV of the hard scattering [4].

6.3 Electrons

6.3.1 Reconstruction

The electron object is reconstructed from an ID track matched to an electromagnetic shower in the calorimeter. Bremsstrahlung photons emitted in the upstream material can spread the shower energy over several clusters and alter the curvature of the electron track. Both effects are treated in the reconstruction [5]:

1. **Topo-cluster reconstruction.** Neighbouring cells are grouped into topological clusters according to their signal significance above the expected noise. Clusters with a substantial electromagnetic energy fraction are used as electron seeds.

2. **Track refitting and matching.** Electron-track candidates are refitted with a Gaussian-sum filter, in which a mixture of Gaussian components is used to describe radiative energy losses. The refitted tracks are extrapolated to the second layer of the EM calorimeter and matched to clusters using their differences in η and ϕ . If several tracks match the same cluster, the ΔR between each extrapolated track position and the cluster centre is used together with the pixel and SCT hits to rank the tracks.
3. **Supercluster formation.** Nearby clusters are added to the seed to recover energy from bremsstrahlung. Additional clusters sharing the matched electron track can be collected over a wider region in ϕ .

The cluster association is illustrated in Figure 6.2. The wider collection in ϕ helps recover energy separated from the electron by its bending in the magnetic field.

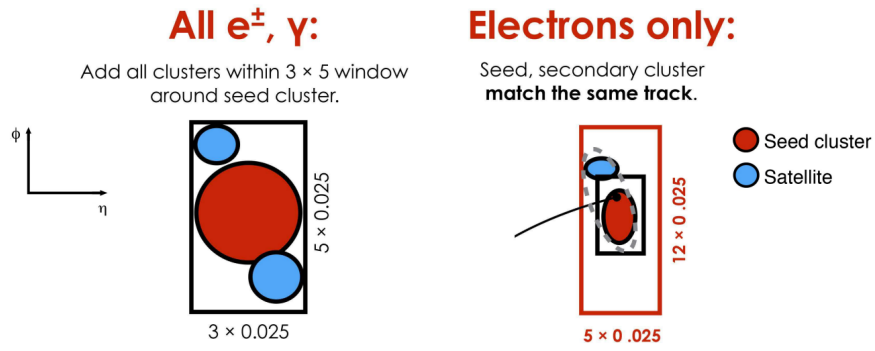


Fig. 6.2 Cluster association for electron superclusters. Seed and additional clusters are shown in red and blue, respectively. Geometrically nearby clusters are collected [5].

6.3.2 Identification

Electron candidates are distinguished from hadronic showers and converted photons using a likelihood discriminant [6]. Its inputs include the shower shapes in the EM calorimeter, track-hit variables, and the agreement between the track and calorimeter measurements. The Loose, Medium and Tight working points are defined and provide increasing background rejection with the different likelihood discriminant threshold. The Tight likelihood working point, denoted TightLH, is used for all electron candidates in this analysis. Its typical prompt-electron identification efficiency is about 80% [7].

6.3.3 Isolation

Electrons from hadron decays can pass the electron identification requirements. These electrons are usually accompanied by other particles from the same jet, while prompt electrons typically have smaller nearby track p_T and calorimeter E_T sums. Isolation requirements are therefore used to suppress this background.

Two isolation variables are calculated using tracks and calorimeter clusters within cones around the electron direction [5]:

- **Track isolation** is performed using the p_T sum of tracks in the cone. At high electron p_T , other decay products can lie close to the electron due to the boost effect, so the cone radius is reduced.
- **Calorimeter isolation** follows the same principle as track isolation, but uses cluster E_T within a fixed cone of $\Delta R = 0.2$. The corrected sum estimates the E_T from other particles near the electron [5].

In the analysis, baseline electrons are required to pass Loose_VarRad isolation while signal electrons must additionally satisfy HighPtCaloOnly. The Loose_VarRad working point requires the track and calorimeter isolation sums to be below 15% and 20% of the electron p_T , respectively. The HighPtCaloOnly working point requires the calorimeter isolation sum to be below $\max(0.015 p_T, 3.5 \text{ GeV})$ [5]. At high E_T , the prompt-electron efficiencies are about 99% for Loose_VarRad and 90–95% for HighPtCaloOnly [7].

6.3.4 Calibration

The electron energy is calibrated for losses in upstream material, energy outside the cluster and variations in detector response [5]. $Z \rightarrow ee$ events are used for the calibration. Reconstruction, identification and isolation efficiencies are measured with the tag-and-probe method: one electron (the tag) is required to pass strict identification cuts, while the other is used as the probe. After background subtraction, the efficiency is the fraction of probes that pass the selection. For a selection efficiency ε , the scale factor is defined as

$$\text{SF} = \frac{\varepsilon_{\text{data}}}{\varepsilon_{\text{MC}}}. \quad (6.1)$$

The scale factors are then applied to simulated event weights for the electron working points used in this analysis.

6.4 Muons

6.4.1 Reconstruction

Muons are reconstructed using tracks in the ID and MS, together with their energy deposits in the calorimeters [2]. The main reconstruction categories are:

- **Combined muons** are reconstructed by matching and fitting ID and MS tracks, including the energy loss between the two tracking systems.
- **Segment-tagged muons** are reconstructed by matching an ID track to one or more local track segments in the MS when a complete MS track is unavailable.
- **Calorimeter-tagged muons** are reconstructed by matching an ID track to calorimeter deposits compatible with a minimum-ionising particle. Acceptance is recovered in regions with limited MS coverage.
- **MS-extrapolated muons** are reconstructed by extrapolating an MS track towards the interaction region without a matched ID track.

Combined muons are used in this analysis.

6.4.2 Identification

Muon identification tests whether the detector hits and the ID and MS momentum measurements are consistent with a single muon trajectory [2]. Decays of charged mesons in flight can change the trajectory and momentum, leading to a disagreement between the momentum measurements. For muon candidates, this agreement is quantified by the q/p significance,

$$S_{q/p} = \frac{|(q/p)_{\text{ID}} - (q/p)_{\text{MS}}|}{\sqrt{\sigma_{\text{ID}}^2 + \sigma_{\text{MS}}^2}}, \quad (6.2)$$

where $(q/p)_{\text{ID}}$ and $(q/p)_{\text{MS}}$ are the charge-to-momentum ratios determined from the ID and MS tracks, respectively. σ_{ID} and σ_{MS} denote the uncertainties. Large values indicate poor agreement and help reject incorrectly matched candidates.

In the analysis, baseline muons are required to pass Loose identification while signal muons must additionally satisfy the Medium working point. The Medium working point requires combined candidates within the ID acceptance to satisfy $S_{q/p} < 7$ and, outside $|\eta| < 0.1$, have at least two MS precision stations with at least three MDT or CSC hits each. The Loose working point applies the same requirements and additionally accepts segment-tagged and

calorimeter-tagged candidates in $|\eta| < 0.1$, where MS coverage is limited [2]. The typical prompt-muon efficiencies are about 99% for Loose and 97% for Medium [2].

6.4.3 Isolation

Muons from heavy-flavour decays are often accompanied by other particles inside a jet. They are suppressed using track and calorimeter isolation, defined by the surrounding activity after excluding the muon's own contribution [2]. Baseline muons are required to pass Loose_VarRad isolation, and signal muons must additionally satisfy Tight_VarRad. Both working points use a track-cone radius that decreases as the muon p_T increases and a fixed calorimeter cone of $\Delta R = 0.2$. The track and calorimeter isolation sums must be below 15% and 30% of the muon p_T for Loose_VarRad, respectively. These limits are reduced to 4% and 15% for Tight_VarRad, allowing less activity around the muon and rejecting more muons produced inside jets [2]. The typical prompt-muon isolation efficiencies are about 95% for Loose_VarRad and 86% for Tight_VarRad [8].

6.4.4 Calibration

The muon momentum scale and resolution are calibrated using resonance decays, e.g., $Z \rightarrow \mu\mu$ and $J/\psi \rightarrow \mu\mu$ [9]. The reconstructed mass distributions constrain biases in the muon momentum measurement. The efficiency scale factors are measured with tag-and-probe samples as for electrons and applied to simulated event weights for the selected working points [2].

6.5 Jets

6.5.1 Reconstruction

Jets are reconstructed from the collimated hadrons produced through parton showering and hadronisation. In this analysis, the jet inputs are built with the particle-flow (PFlow) algorithm, which combines ID track momenta with calorimeter energy measurements [10]. The main steps are:

1. Well-measured tracks are matched to calorimeter topological clusters.
2. The expected calorimeter energy deposited by the matched charged particles is subtracted from the clusters, avoiding double counting when the track momenta are used.
3. The selected tracks and remaining calorimeter contributions are retained as the jet inputs. Vertex association is used to suppress charged-particle contributions from pile-up.

Calorimeter contributions from neutral particles and charged particles without a selected track are retained at the EM scale.

The anti- k_t algorithm then groups the PFlow constituents into jets by repeatedly combining them according to their angular separation and p_T [11]. Low- p_T constituents are preferentially clustered around nearby high- p_T constituents, giving isolated jets approximately circular boundaries with an angular radius set by $R = 0.4$. The jet four-momentum is obtained by summing the constituent four-momenta.

6.5.2 Identification and pile-up rejection

Jets in this analysis are required to satisfy LooseBad cleaning and the neural-network jet-vertex tagger (NNJVT) FixedEffPt working point.

- **Jet cleaning** uses calorimeter pulse shapes and the distribution of energy among calorimeter layers to reject noise and non-collision backgrounds. For example, a jet with $|\eta| < 2$ is rejected if more than 99% of its energy is deposited in a single calorimeter layer. The LooseBad efficiency for collision jets exceeds 99.5% [12].
- **Pile-up rejection** requires the NNJVT score to exceed a threshold that depends on jet p_T and η [13]. Larger scores favour jets whose tracks originate from the primary vertex and carry a substantial fraction of the jet p_T .

The NNJVT requirement is applied within the kinematic range of its calibration.

6.5.3 Calibration

The reconstructed jet four-momentum is calibrated through the following sequence [14]:

1. **Pile-up subtraction.** Additional p_T from pile-up is subtracted using the event p_T density ρ and jet area A , with a correction proportional to ρA . Residual pile-up dependence is then corrected.
2. **Energy and direction calibration.** The mean energy response and η bias are corrected using particle-level jets, built from stable simulated particles excluding muons and neutrinos.
3. **Global sequential calibration.** Variations in the jet response due to particle composition and shower development are corrected using tracking and calorimeter variables.

4. **Residual in situ calibration.** Remaining differences between data and simulation are measured with momentum balance in Z+jet, photon+jet and multijet events. This final correction is applied to collision data.

Jet energy-scale and resolution uncertainties are propagated to the analysis selections and to E_T^{miss} . Scale factors for the pile-up rejection efficiency are applied to simulated event weights.

6.6 Jet flavour tagging

6.6.1 Reconstruction inputs

Jets containing b -hadrons are identified using the displacement of their decay products from the primary vertex. With a typical proper lifetime of about 1.5 ps, a boosted b -hadron can travel several millimetres before decaying [15]. Its charged decay products can therefore have significant impact parameters and a common secondary vertex. The subsequent decay of a charm hadron can produce an additional displaced vertex. The track momenta, impact parameters, detector hit information and correlations between tracks are used as flavour-tagging inputs.

6.6.2 Identification

The GN2v01 algorithm is used for b -tagging in this analysis. Jet and track information is processed with a transformer neural network, in which the representation of each track is updated using information from other tracks in the jet [15]. Tracks with a common displaced origin can thereby be recognised together. During training, auxiliary tasks are used to identify track origins and to group tracks from the same vertex, as illustrated in Figure 6.3. Explicitly reconstructed secondary vertices are not required as separate input objects.

The outputs for the b -, c -, hadronic-tau and light-jet hypotheses are combined into the discriminant

$$D_b = \ln \left[\frac{p_b}{f_c p_c + f_\tau p_\tau + (1 - f_c - f_\tau) p_u} \right], \quad (6.3)$$

where p_b , p_c , p_τ and p_u are the classifier outputs [15]. The fixed coefficients f_c and f_τ set the balance between the background flavours in the discriminant. The GN2v01 85% working point is used in this analysis. A jet is tagged when D_b exceeds a fixed threshold chosen to retain 85% of b -jets in simulated $t\bar{t}$ events. The efficiency for signal b -jets can differ because of their different kinematic distributions.

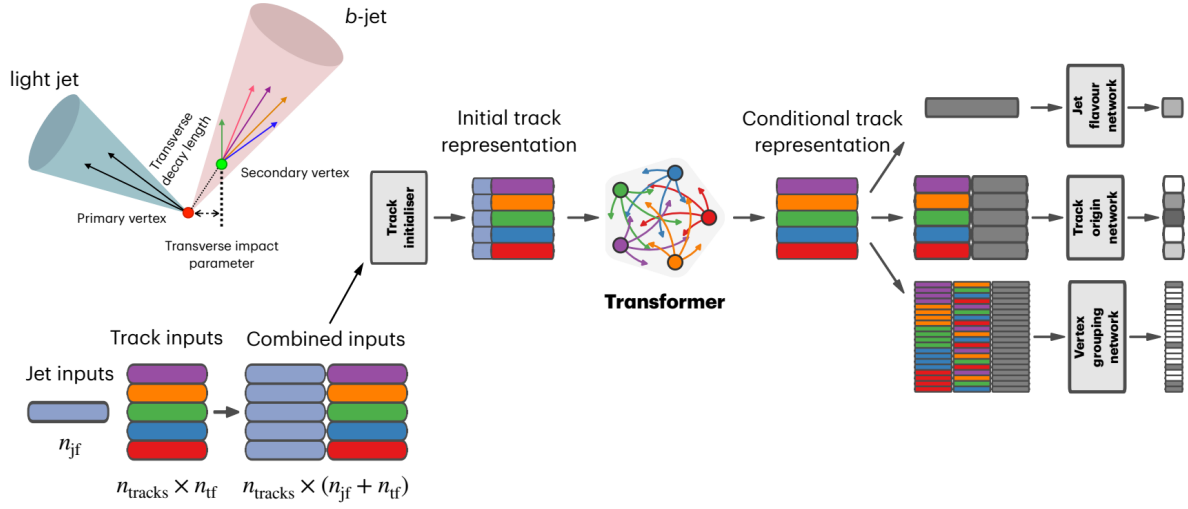


Fig. 6.3 GN2 flavour-tagging architecture. Jet and track measurements are combined, processed by the transformer, and used for jet-flavour prediction and the auxiliary track-origin and vertex-grouping tasks. The symbols n_{jf} , n_{tf} and n_{tracks} denote the numbers of jet features, track features and tracks [15].

6.6.3 Calibration

Tagging and mistagging efficiencies are measured in flavour-enriched data samples and compared with simulation [15]. Flavour- and kinematics-dependent scale factors are applied to simulated event weights. Both tagged and untagged outcomes must be corrected. For a single jet with tagging efficiency ϵ_f , the correction for an untagged outcome is

$$w_f^{\text{untagged}} = \frac{1 - \epsilon_f^{\text{data}}}{1 - \epsilon_f^{\text{MC}}} = \frac{1 - \text{SF}_f \epsilon_f^{\text{MC}}}{1 - \epsilon_f^{\text{MC}}}. \quad (6.4)$$

This complementary-probability relation explains why flavour-tagging corrections affect both the zero-tag and tagged event categories. In this search, the zero-tag category also receives a direct contribution from $U_1 \rightarrow s\tau$ when β_L^{23} is sizeable, as discussed in Chapter 7.

6.7 Hadronic tau leptons

6.7.1 Reconstruction

τ leptons can decay both leptonically and hadronically. Approximately 65% of τ decays are hadronic, with the major modes containing charged and possible neutral pions [16]. In this

analysis we only consider the hadronic decay tau. It is denoted by τ_{had} . Here the τ -neutrino is excluded in the reconstruction.

The hadronic tau candidates are seeded by anti- k_t jets with $R = 0.4$. The production vertex is selected using tracks within $\Delta R < 0.2$ of the seed axis, choosing the vertex with the largest fraction of their p_T sum. Associated tracks are classified by a recurrent neural network (RNN) to select tau-decay tracks. Candidates with one or three tau-decay tracks are called one-prong or three-prong candidates.

6.7.2 Identification

Hadronic tau candidates are distinguished from quark- and gluon-initiated jets using a separate RNN discriminant [17, 16]. Its inputs include track momenta and displacements, calorimeter shower properties and, for three-prong candidates, the significance of the reconstructed decay-vertex displacement. These variables exploit the narrow shower, low charged multiplicity and displaced decay of a tau lepton. Nearby activity is included, without a separate isolation working point.

Baseline tau candidates are required to pass Loose RNN identification, and signal candidates must additionally satisfy Tight. The higher RNN-score threshold for Tight rejects more jets at the cost of tau efficiency. For reconstructed one-prong (three-prong) candidates, the target identification efficiencies are 85% (75%) for Loose and 60% (45%) for Tight [16].

Both baseline and signal tau candidates must also pass Loose electron rejection. This working point requires a minimum score from a separate RNN using shower and track information, including the relation between calorimeter energy and track momentum. The tau efficiencies are approximately 95% and 98% for one-prong and three-prong candidates, respectively [16]. Muon rejection uses the angular separation from reconstructed muons.

6.7.3 Calibration

The tau energy scale is calibrated to the visible decay products [16]. A calorimeter estimate, corrected for pile-up and detector response, is combined with an estimate from charged-particle tracks and neutral calorimeter deposits, using weights determined by their expected resolutions. A boosted regression tree then corrects the combined estimate using shower properties, track multiplicity and reconstructed decay structure. This calibration is used for the tau kinematics in the analysis.

Simulated event weights are corrected for differences in tau reconstruction, identification and electron-rejection efficiencies between data and simulation. For signal candidates pass-

ing Tight tau identification, the electron-rejection scale factor associated with Medium tau identification is used.

6.8 Overlap removal

6.8.1 Shared signals and matching

The same detector activity can be reconstructed as more than one object. For example, an electron shower can also form a jet, while a hadronic tau candidate is seeded by a calorimeter jet. An ordered overlap-removal procedure is applied to the baseline candidates to resolve these ambiguities before event selection. Only objects surviving an earlier step are considered in subsequent steps.

Geometrical overlaps are evaluated with the rapidity distance

$$\Delta R_y = \sqrt{(y_i - y_j)^2 + (\phi_i - \phi_j)^2}, \quad y = \frac{1}{2} \ln \frac{E + p_z}{E - p_z}, \quad (6.5)$$

where y is rapidity. In the massless limit, $y = \eta$ and $\Delta R_y = \Delta R$. Shared ID tracks are also used to identify duplicate electron and muon interpretations. For muon–jet matching, *ghost association* is used: an object with negligible momentum is placed along the muon direction during jet clustering, and the jet containing it is identified. The association is therefore determined by the clustering without changing the jet momentum.

6.8.2 Removal sequence

The following steps are applied in order:

1. **Electron–electron:** when two electrons share an ID track, the electron with lower p_T is removed.
2. **Tau–electron:** a tau candidate within $\Delta R_y < 0.2$ of an electron is removed.
3. **Tau–muon:** a tau candidate within $\Delta R_y < 0.2$ of a muon is removed.
4. **Calorimeter-tagged muon–electron:** a calorimeter-tagged muon sharing the electron ID track is removed.
5. **Electron–muon:** an electron sharing an ID track with a remaining muon is removed.
6. **Jet–electron:** a jet within $\Delta R_y < 0.2$ of an electron is removed.

7. **Electron–jet**: an electron within $\Delta R_y < 0.4$ of a remaining jet is removed.
8. **Jet–muon**: a jet with fewer than three associated tracks is removed if the muon is ghost-associated with it.
9. **Muon–jet**: a muon within $\Delta R_y < 0.4$ of a remaining jet is removed.
10. **Jet–tau**: a jet within $\Delta R_y < 0.2$ of a remaining tau candidate is removed.

The inner electron–jet step retains the electron interpretation of an electromagnetic shower, whereas the outer step rejects electrons embedded in surviving hadronic jets. For muon–jet overlaps, the small track multiplicity is used to recognise a jet candidate dominated by activity associated with the muon. The last step retains the tau interpretation of an overlapping jet. The separate low- p_T muon rejection in tau identification, described in Section 6.7, is applied in addition to this baseline-object sequence.

6.9 Missing transverse momentum

6.9.1 Reconstruction

The incoming partons carry little transverse momentum compared with the hard-scattering scale, so an imbalance in the measured transverse momentum can indicate undetected particles. Both the neutrino from the hard process and the tau-decay neutrino contribute to the signal imbalance. The reconstructed vector is obtained from the negative sum of calibrated visible contributions [18]:

$$\vec{p}_T^{\text{miss}} = - \left(\sum_e \vec{p}_T^e + \sum_\mu \vec{p}_T^\mu + \sum_{\tau_{\text{had}}} \vec{p}_T^\tau + \sum_j \vec{p}_T^j + \vec{p}_T^{\text{soft}} \right), \quad (6.6)$$

$$E_T^{\text{miss}} = \left| \vec{p}_T^{\text{miss}} \right|. \quad (6.7)$$

Two types of contribution are included:

- **Hard term**: the electron, muon, hadronic-tau and jet contributions accepted by the reconstruction. No separate photon collection is supplied in this analysis.
- **Track soft term**: the vector sum of selected tracks associated with the primary vertex and not assigned to retained hard-object contributions.

The primary-vertex requirement suppresses charged-particle contributions from pile-up. Neutral particles outside the hard term are not measured by the track soft term.

The baseline electron, muon and tau identification and isolation working points are used to select the corresponding inputs. Jets are selected separately by the E_T^{miss} reconstruction with the Tight operating point. This requires jet $p_T > 20$ GeV for $|\eta| < 2.5$ and $p_T > 30$ GeV for $2.5 \leq |\eta| < 4.5$. The higher forward threshold suppresses pile-up jets beyond the tracking acceptance [18]. Their acceptance is distinct from that of the central jets counted in the analysis regions. An association map is used to record shared detector signals and resolve double counting when the object contributions are combined [18]. Consequently, E_T^{miss} is not recomputed as a simple sum of the objects passing the final region selections or the geometrical overlap-removal procedure.

6.9.2 Calibration and uncertainties

The energy and momentum calibrations of the hard objects are propagated to the reconstructed vector. The response and resolution uncertainties of the track soft term are evaluated separately using events in which little genuine E_T^{miss} is expected [18]. The corresponding variations are propagated together with the hard-object uncertainties to retain their correlations with the visible kinematics.

Apparent E_T^{miss} can also be produced by mismeasured visible momentum, calorimeter noise or energy lost in uninstrumented regions. For example, an underestimated jet momentum can produce an imbalance along the jet direction. Jet cleaning and event-level angular selections are therefore used to suppress these contributions, as described in Chapter 7.

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